

PETER ZENG

PHD CANDIDATE IN COMPUTER SCIENCE • STONY BROOK, NY

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EDUCATION

Stony Brook University

Ph.D. in Computer Science

August 2023 - Expected May 2027

Advisor: [Owen Rambow](#)

Stony Brook University

BS/MS in Computer Science

August 2017 - May 2022

RESEARCH INTERESTS

- **Machine Theory of Mind & Cognitively Informed AI:** Assessing the ability of Large Language Models (LLMs) and Vision-Language Models (LVLMs) to model belief states, track recursive mental states, and perform human-like cognitive reasoning.
- **Multimodal Grounding & Human-AI Communication:** Investigating how LVLMs ground language to vision and coordinate with human partners, with a focus on divergences between human pragmatic inference and model behavior in interactive settings.
- **AI Interpretability & Trustworthiness:** Developing hybrid architectures that combine the performance of deep neural networks with the transparency of feature-based models (e.g., residualized similarity).

SELECTED PUBLICATIONS & MANUSCRIPTS

Implicit vs. Explicit Prompting Strategies for LVLMs in Referential Communication

Spring 2026

**Peter Zeng, Amie J. Paige, Weiling Li, Susan E. Brennan, Owen Rambow, Cameron R. Jones. Under Review.*

- **Problem:** Two prior studies reached contradictory conclusions about whether LVLMs can coordinate efficient referring expressions, leaving it unclear whether task design or prompting approach was responsible.
- **Method:** Controlled for task differences between the conflicting studies while directly comparing explicit versus implicit prompting strategies across models, isolating prompting as the key variable.
- **Outcome:** Showed that LVLMs succeed at communicative efficiency when explicitly instructed to do so, but fail to infer this need from implicit prompts — unlike humans, who derive communicative intent from context alone — revealing a fundamental gap in pragmatic inference.

LVLMs and Humans Ground Differently in Referential Communication

Fall 2025

**Peter Zeng, Weiling Li, Amie J. Paige, Zhengxiang Wang, Panagiotis Kaliosis,*

Dimitris Samaras, Gregory Zelinsky, Susan E. Brennan, and Owen Rambow. ACL 2026 (Main).

- **Problem:** LVLMs appear competent in multimodal reasoning, but it's unclear whether they ground the same way humans do during interactive communication.
- **Method:** Designed and implemented a referential communication task comparing human-human and human-LVLM interactions. Analyzed linguistic strategies, visual attention patterns, and error profiles across conditions to isolate model-human divergences.
- **Outcome:** Demonstrated that LVLMs systematically rely on surface-level visual heuristics and fail to adapt grounding strategies to partner feedback, unlike humans who use pragmatic inference and incremental clarification. Findings reveal a fundamental mismatch between human and model grounding processes and establish a new evaluation paradigm for interactive multimodal communication.

Residualized Similarity for Interpretability in Authorship Verification

Spring 2025

**Peter Zeng, Pegah Alipoormolabashi, Jihu Mun, Gourab Dey, Nikita Soni,*

Niranjana Balasubramanian, Owen Rambow, H. Schwartz. EMNLP 2025 (Findings).

- **Problem:** Standard cosine-similarity models in authorship attribution often sacrifice interpretability for performance.
- **Method:** Developed a novel neural framework that predicts the "residual" (error) of an interpretable baseline, allowing the model to capture complex non-linearities while retaining feature-level explainability.
- **Outcome:** Improved accuracy by ~20-30% over baseline interpretable systems, establishing a new SOTA for faithful explainability in authorship tasks.

Gram2Vec: An Interpretable Document Vectorizer

**Peter Zeng, Hannah Stortz, Eric Sclafani, Alina Shabaeva,
Maria Elizabeth Garza, Daniel Greeson, Owen Rambow. Under Review.*

- **Problem:** Neural text embeddings are often "black boxes," making it difficult to understand the linguistic basis for authorship attribution.
- **Method:** Created a grammatical style embedding algorithm that extracts normalized relative frequencies of morphosyntactic features into a higher-dimensional space.
- **Outcome:** Demonstrated that explicit grammatical feature extraction can be successfully applied to two domains: authorship verification and AI detection.

Synthetic Audio Helps for Cognitive State Tasks

Fall 2024

*Adil Soubki, John Murzaku, *Peter Zeng, Owen Rambow. NAACL 2025 (Findings).*

- **Problem:** Gold-standard audio data for cognitive tasks is scarce, yet text-only models miss prosodic cues.
- **Method:** Introduced Synthetic Audio Data fine-tuning (SAD), a multimodal training approach using text and zero-shot synthetic audio from off-the-shelf TTS systems.
- **Outcome:** Proved that synthetic audio acts as an effective regularizer, improving cognitive state modeling performance even when real audio is unavailable.

Views Are My Own, But Also Yours: Benchmarking Theory of Mind

Fall 2023

*Adil Soubki, John Murzaku, Arash Yousefi Jordehi, *Peter Zeng,
Magdalena Markowska, Seyed Abolghasem Mirroshandel, Owen Rambow. ACL 2024 (Findings).*

- **Problem:** Existing Theory of Mind benchmarks rely on synthetic scenarios that do not reflect the complexity of natural dialogue.
- **Method:** Constructed *Common-ToM*, a benchmark derived from naturally occurring dialogs, to test LLM capabilities zero-shot.
- **Outcome:** Demonstrated that while LLMs retrieve facts well, they struggle to track recursive belief states without explicit belief representations.

Re-Examining FactBank: Predicting the Author's Presentation of Factuality

Spring 2022

*John Murzaku, *Peter Zeng, Magdalena Markowska, Owen Rambow. COLING 2022.*

- **Problem:** Prior research on the canonical FactBank corpus relied on data representation errors, leading to invalid conclusions.
- **Method:** Systematically identified and corrected representation errors in the corpus to create a reliable ground truth.
- **Outcome:** Achieved State-of-the-Art results for predicting authorial factuality across four distinct corpora using the corrected data.

PROFESSIONAL EXPERIENCE

Applied Scientist Intern

June 2026 - September 2026

Adobe (AEP Agent Orchestrator)

San Jose, CA

- Designing and implementing memory management systems for the Adobe Experience Platform (AEP) Agent Orchestrator, enabling agents to persist, retrieve, and reason over long-horizon context.
- Researching memory architectures (episodic, semantic, and working memory) to improve agent coherence and task continuity across multi-step orchestration workflows.
- Collaborating with cross-functional teams to evaluate memory retrieval strategies and their downstream impact on agent response quality and latency.

Research Assistant

June 2025 - Present

BIAS-NRT (NSF Fellow)

Stony Brook, NY

Detecting and Addressing Bias in Data, Humans, & Institutions

- Conducting task analyses with stakeholders at an Innocence Organization to define user needs and iteratively design optimal prompts for CoCounsel (a secure LLM) to assist in the intake process.
- Developing multimodal evaluation frameworks to assess how Vision-Language Models (VLMs) describe and match objects compared to human baselines.
- Investigating bias propagation in large-scale datasets used for training generative models.

Research Assistant

August 2023 - May 2025

IARPA HIATUS (Stony Brook University)

Stony Brook, NY

- Lead researcher on the "Residualized Similarity" project, focusing on attribution in low-resource and obfuscated text domains.
- Managed data pipelines for large-scale authorship corpora, ensuring rigorous train/test separation to prevent data leakage in open-world attribution tasks.

Software Engineer

June 2022 - August 2023

Amazon Web Services (Elastic Container Service)

New York, NY

- **High-Scale Infrastructure:** Maintained core control-plane infrastructure for Amazon ECS, supporting 2 million daily customer deployments across global regions.
- **Reliability Engineering:** Designed and implemented alarm-based rollback mechanisms, requiring the ingestion and real-time processing of massive-scale telemetry data to ensure service availability.
- **Production Excellence:** Participated in on-call rotations for a Tier-1 AWS service, resolving critical production incidents affecting enterprise customers.

TECHNICAL SKILLS

Languages	Python, Java
Deep Learning Libraries	PyTorch, HuggingFace (Transformers, PEFT), W&B
AI Techniques	Fine-Tuning, Prompt Engineering, RAG
Infrastructure	Git, Linux/Unix
Methodologies	Causal Inference, Human-Subjects Evaluation, Survey Design

ACADEMIC SERVICE & TEACHING

- **Reviewer:** ACL Rolling Review (ARR), NeurIPS 2026 Evaluations and Datasets Track
- **Teaching Assistant (Stony Brook):** CSE 385 Analysis of Algorithms, CSE 160/260 OOP & Data Structures
- **Academic Tutor (ASTC):** CSE 101 (Intro to Programming) & PHY 131/132 (Physics Mechanics) at the *Academic Success and Tutoring Center*, Stony Brook, NY
- **Instructor:** Taught K-12 introductory programming (Scratch, Python, Java) at *theCoderSchool*, Buffalo, NY